

Deep Learning Portfolio Examination

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Part - 1: Stabilizing the Existing Codebase

Performance Report

Each of the four recovered diagnostic image profiles (cells, chest, lesions, orgs) was benchmarked against all three architectures (AlexNet, VGG16, ResNet18) using the reconstructed, config-driven pipeline. The complete per-epoch training and validation logs and the final held-out test metrics are reproduced below for every dataset-model permutation.

Dataset Used – cells

Blood cell microscopy (8 classes, RGB) | Expected minimum test accuracy: 90%

1. cells – AlexNet

Total epochs trained: 15

Configuration Used:

```
ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 3
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 8
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.9301 - Train Acc: 63.56% | Val Loss: 3.1397 - Val Acc: 36.36%
Epoch [02/15] | Train Loss: 0.4248 - Train Acc: 84.58% | Val Loss: 0.9293 - Val Acc: 68.25%
Epoch [03/15] | Train Loss: 0.3071 - Train Acc: 89.87% | Val Loss: 0.3447 - Val Acc: 88.59%
Epoch [04/15] | Train Loss: 0.2565 - Train Acc: 91.68% | Val Loss: 0.5073 - Val Acc: 79.81%
Epoch [05/15] | Train Loss: 0.2163 - Train Acc: 93.25% | Val Loss: 2.3361 - Val Acc: 57.94%
Epoch [06/15] | Train Loss: 0.1875 - Train Acc: 93.78% | Val Loss: 0.4023 - Val Acc: 86.32%
Epoch [07/15] | Train Loss: 0.1736 - Train Acc: 94.26% | Val Loss: 1.6407 - Val Acc: 64.01%
Epoch [08/15] | Train Loss: 0.1862 - Train Acc: 94.21% | Val Loss: 0.8748 - Val Acc: 69.86%
Epoch [09/15] | Train Loss: 0.1644 - Train Acc: 94.78% | Val Loss: 0.2744 - Val Acc: 89.83%
Epoch [10/15] | Train Loss: 0.1454 - Train Acc: 95.27% | Val Loss: 0.2365 - Val Acc: 92.90%
Epoch [11/15] | Train Loss: 0.1482 - Train Acc: 95.17% | Val Loss: 0.2147 - Val Acc: 93.12%
Epoch [12/15] | Train Loss: 0.1422 - Train Acc: 95.55% | Val Loss: 0.2316 - Val Acc: 92.47%
Epoch [13/15] | Train Loss: 0.1256 - Train Acc: 95.94% | Val Loss: 0.3447 - Val Acc: 91.15%
Epoch [14/15] | Train Loss: 0.1046 - Train Acc: 96.76% | Val Loss: 0.2586 - Val Acc: 92.90%
Epoch [15/15] | Train Loss: 0.0985 - Train Acc: 96.85% | Val Loss: 0.2213 - Val Acc: 93.64%
Best checkpoint saved: best_model/cells_AlexNet.pth (epoch=15, val_acc=93.64%)
Training Complete!
```

Final Test Results (AlexNet on cells): Accuracy = 95.15% | Precision (macro) = 0.9474 | Recall (macro) = 0.9443 | Macro F1 = 0.9445

2. cells – VGG16

Total epochs trained: 15

Configuration Used:

```
ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 3
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 8
OPTIMIZER..... Adam
```

```
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.7807 - Train Acc: 70.35% | Val Loss: 0.5020 - Val Acc: 81.57%
Epoch [02/15] | Train Loss: 0.4778 - Train Acc: 83.65% | Val Loss: 0.7367 - Val Acc: 78.05%
Epoch [03/15] | Train Loss: 0.3864 - Train Acc: 87.65% | Val Loss: 0.6123 - Val Acc: 77.76%
Epoch [04/15] | Train Loss: 0.3115 - Train Acc: 90.32% | Val Loss: 0.6647 - Val Acc: 75.20%
Epoch [05/15] | Train Loss: 0.3170 - Train Acc: 90.10% | Val Loss: 0.6265 - Val Acc: 80.03%
Epoch [06/15] | Train Loss: 0.2479 - Train Acc: 92.34% | Val Loss: 0.3569 - Val Acc: 89.39%
Epoch [07/15] | Train Loss: 0.2155 - Train Acc: 93.08% | Val Loss: 0.2055 - Val Acc: 93.93%
Epoch [08/15] | Train Loss: 0.2004 - Train Acc: 93.95% | Val Loss: 0.4448 - Val Acc: 86.83%
Epoch [09/15] | Train Loss: 0.1843 - Train Acc: 94.57% | Val Loss: 0.2220 - Val Acc: 93.05%
Epoch [10/15] | Train Loss: 0.1573 - Train Acc: 95.13% | Val Loss: 0.1483 - Val Acc: 95.68%
Epoch [11/15] | Train Loss: 0.1366 - Train Acc: 95.77% | Val Loss: 0.3248 - Val Acc: 88.66%
Epoch [12/15] | Train Loss: 0.1574 - Train Acc: 95.20% | Val Loss: 0.1593 - Val Acc: 94.51%
Epoch [13/15] | Train Loss: 0.1475 - Train Acc: 95.42% | Val Loss: 0.1970 - Val Acc: 93.34%
Epoch [14/15] | Train Loss: 0.1287 - Train Acc: 95.91% | Val Loss: 0.1637 - Val Acc: 95.25%
Epoch [15/15] | Train Loss: 0.1112 - Train Acc: 96.65% | Val Loss: 0.1890 - Val Acc: 94.15%
Best checkpoint saved: best_model/cells_VGG16.pth (epoch=10, val_acc=95.68%)
Training Complete!
```

Final Test Results (VGG16 on cells): Accuracy = 96.78% | Precision (macro) = 0.9646 | Recall (macro) = 0.9685 | Macro F1 = 0.9665

3. cells – ResNet18

Total epochs trained: 15

Configuration Used:

```
ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 3
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 8
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.4373 - Train Acc: 84.42% | Val Loss: 0.7451 - Val Acc: 79.44%
Epoch [02/15] | Train Loss: 0.2135 - Train Acc: 92.99% | Val Loss: 1.0535 - Val Acc: 73.45%
Epoch [03/15] | Train Loss: 0.1472 - Train Acc: 95.23% | Val Loss: 0.4059 - Val Acc: 85.52%
Epoch [04/15] | Train Loss: 0.1450 - Train Acc: 95.02% | Val Loss: 0.2614 - Val Acc: 90.71%
Epoch [05/15] | Train Loss: 0.1137 - Train Acc: 96.25% | Val Loss: 0.7099 - Val Acc: 80.61%
Epoch [06/15] | Train Loss: 0.1011 - Train Acc: 96.54% | Val Loss: 0.2834 - Val Acc: 90.86%
Epoch [07/15] | Train Loss: 0.1131 - Train Acc: 96.21% | Val Loss: 0.2845 - Val Acc: 90.56%
Epoch [08/15] | Train Loss: 0.0874 - Train Acc: 97.11% | Val Loss: 0.3889 - Val Acc: 88.88%
Epoch [09/15] | Train Loss: 0.0906 - Train Acc: 96.68% | Val Loss: 0.4331 - Val Acc: 86.47%
Epoch [10/15] | Train Loss: 0.0812 - Train Acc: 97.24% | Val Loss: 0.3649 - Val Acc: 88.95%
Epoch [11/15] | Train Loss: 0.0669 - Train Acc: 97.71% | Val Loss: 0.5524 - Val Acc: 86.10%
Epoch [12/15] | Train Loss: 0.0813 - Train Acc: 97.22% | Val Loss: 0.5019 - Val Acc: 85.81%
Epoch [13/15] | Train Loss: 0.0628 - Train Acc: 97.85% | Val Loss: 0.6183 - Val Acc: 85.59%
Epoch [14/15] | Train Loss: 0.0668 - Train Acc: 97.74% | Val Loss: 0.2208 - Val Acc: 93.20%
Epoch [15/15] | Train Loss: 0.0569 - Train Acc: 98.07% | Val Loss: 0.1266 - Val Acc: 96.12%
Best checkpoint saved: best_model/cells_ResNet18.pth (epoch=15, val_acc=96.12%)
Training Complete!
```

Final Test Results (ResNet18 on cells): Accuracy = 96.35% | Precision (macro) = 0.9599 | Recall (macro) = 0.9641 | Macro F1 = 0.9613

Dataset Used – chest

Chest X-ray (2 classes, grayscale) | Expected minimum test accuracy: 87%

4. chest – AlexNet

Total epochs trained: 15

Configuration Used:

```
ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 1
```

CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 2
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 1e-4

Logs:

Epoch [01/15]		Train Loss: 0.3124	-	Train Acc: 87.15%		Val Loss: 0.2966	-	Val Acc: 91.97%
Epoch [02/15]		Train Loss: 0.1586	-	Train Acc: 94.20%		Val Loss: 0.2135	-	Val Acc: 94.84%
Epoch [03/15]		Train Loss: 0.1320	-	Train Acc: 94.84%		Val Loss: 0.1117	-	Val Acc: 95.22%
Epoch [04/15]		Train Loss: 0.1228	-	Train Acc: 95.63%		Val Loss: 0.1242	-	Val Acc: 95.79%
Epoch [05/15]		Train Loss: 0.0795	-	Train Acc: 97.01%		Val Loss: 0.4370	-	Val Acc: 86.04%
Epoch [06/15]		Train Loss: 0.0965	-	Train Acc: 96.56%		Val Loss: 0.3662	-	Val Acc: 86.62%
Epoch [07/15]		Train Loss: 0.0977	-	Train Acc: 96.58%		Val Loss: 0.1532	-	Val Acc: 95.03%
Epoch [08/15]		Train Loss: 0.0780	-	Train Acc: 97.41%		Val Loss: 0.1836	-	Val Acc: 95.79%
Epoch [09/15]		Train Loss: 0.0614	-	Train Acc: 97.88%		Val Loss: 0.1328	-	Val Acc: 95.98%
Epoch [10/15]		Train Loss: 0.0595	-	Train Acc: 97.90%		Val Loss: 0.1617	-	Val Acc: 95.03%
Epoch [11/15]		Train Loss: 0.0482	-	Train Acc: 98.19%		Val Loss: 0.1559	-	Val Acc: 95.22%
Epoch [12/15]		Train Loss: 0.0465	-	Train Acc: 98.45%		Val Loss: 0.0932	-	Val Acc: 96.75%
Epoch [13/15]		Train Loss: 0.0471	-	Train Acc: 98.51%		Val Loss: 0.1422	-	Val Acc: 95.98%
Epoch [14/15]		Train Loss: 0.0393	-	Train Acc: 98.47%		Val Loss: 0.0817	-	Val Acc: 97.90%
Epoch [15/15]		Train Loss: 0.0348	-	Train Acc: 98.92%		Val Loss: 0.1391	-	Val Acc: 95.22%

Best checkpoint saved: best_model/chest_AlexNet.pth (epoch=14, val_acc=97.90%)
Training Complete!

Final Test Results (AlexNet on chest): Accuracy = 86.38% | Precision (macro) = 0.9054 | Recall (macro) = 0.8201 | Macro F1 = 0.8408

5. chest – VGG16

Total epochs trained: 15

Configuration Used:

ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 1
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 2
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 1e-4

Logs:

Epoch [01/15]		Train Loss: 0.2293	-	Train Acc: 90.59%		Val Loss: 0.3932	-	Val Acc: 85.85%
Epoch [02/15]		Train Loss: 0.1342	-	Train Acc: 95.29%		Val Loss: 0.5343	-	Val Acc: 83.17%
Epoch [03/15]		Train Loss: 0.0983	-	Train Acc: 96.45%		Val Loss: 0.1144	-	Val Acc: 95.22%
Epoch [04/15]		Train Loss: 0.0907	-	Train Acc: 96.58%		Val Loss: 0.1326	-	Val Acc: 95.60%
Epoch [05/15]		Train Loss: 0.0853	-	Train Acc: 96.62%		Val Loss: 1.8916	-	Val Acc: 61.19%
Epoch [06/15]		Train Loss: 0.1042	-	Train Acc: 96.50%		Val Loss: 0.1755	-	Val Acc: 95.98%
Epoch [07/15]		Train Loss: 0.0813	-	Train Acc: 97.15%		Val Loss: 0.1425	-	Val Acc: 94.65%
Epoch [08/15]		Train Loss: 0.0695	-	Train Acc: 97.49%		Val Loss: 0.0914	-	Val Acc: 95.60%
Epoch [09/15]		Train Loss: 0.0548	-	Train Acc: 97.98%		Val Loss: 0.2228	-	Val Acc: 95.03%
Epoch [10/15]		Train Loss: 0.0496	-	Train Acc: 98.15%		Val Loss: 0.0964	-	Val Acc: 95.98%
Epoch [11/15]		Train Loss: 0.0610	-	Train Acc: 97.90%		Val Loss: 0.3047	-	Val Acc: 91.20%
Epoch [12/15]		Train Loss: 0.0487	-	Train Acc: 98.13%		Val Loss: 0.1201	-	Val Acc: 96.18%
Epoch [13/15]		Train Loss: 0.0488	-	Train Acc: 98.24%		Val Loss: 0.1335	-	Val Acc: 95.98%
Epoch [14/15]		Train Loss: 0.0437	-	Train Acc: 98.81%		Val Loss: 0.2581	-	Val Acc: 95.98%
Epoch [15/15]		Train Loss: 0.0345	-	Train Acc: 98.68%		Val Loss: 0.0953	-	Val Acc: 96.94%

Best checkpoint saved: best_model/chest_VGG16.pth (epoch=15, val_acc=96.94%)
Training Complete!

Final Test Results (VGG16 on chest): Accuracy = 82.21% | Precision (macro) = 0.8892 | Recall (macro) = 0.7628 | Macro F1 = 0.7822

6. chest – ResNet18

Total epochs trained: 15

Configuration Used:

ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 1
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 2
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 1e-4

Logs:

Epoch [01/15]		Train Loss: 0.2396	-	Train Acc: 90.25%		Val Loss: 0.3194	-	Val Acc: 87.95%
Epoch [02/15]		Train Loss: 0.1516	-	Train Acc: 94.18%		Val Loss: 0.4163	-	Val Acc: 86.81%
Epoch [03/15]		Train Loss: 0.1176	-	Train Acc: 95.52%		Val Loss: 0.2554	-	Val Acc: 91.59%
Epoch [04/15]		Train Loss: 0.1124	-	Train Acc: 95.82%		Val Loss: 0.1806	-	Val Acc: 93.69%
Epoch [05/15]		Train Loss: 0.0987	-	Train Acc: 96.24%		Val Loss: 1.6548	-	Val Acc: 65.97%
Epoch [06/15]		Train Loss: 0.0926	-	Train Acc: 96.64%		Val Loss: 0.2739	-	Val Acc: 91.01%
Epoch [07/15]		Train Loss: 0.0781	-	Train Acc: 97.11%		Val Loss: 0.1037	-	Val Acc: 95.98%
Epoch [08/15]		Train Loss: 0.0702	-	Train Acc: 97.52%		Val Loss: 0.1801	-	Val Acc: 93.69%
Epoch [09/15]		Train Loss: 0.0729	-	Train Acc: 97.09%		Val Loss: 0.1435	-	Val Acc: 93.88%
Epoch [10/15]		Train Loss: 0.0638	-	Train Acc: 97.83%		Val Loss: 0.1197	-	Val Acc: 95.79%
Epoch [11/15]		Train Loss: 0.0601	-	Train Acc: 97.81%		Val Loss: 0.2391	-	Val Acc: 92.35%
Epoch [12/15]		Train Loss: 0.0569	-	Train Acc: 97.81%		Val Loss: 0.1293	-	Val Acc: 95.60%
Epoch [13/15]		Train Loss: 0.0540	-	Train Acc: 98.00%		Val Loss: 0.2655	-	Val Acc: 91.59%
Epoch [14/15]		Train Loss: 0.0507	-	Train Acc: 98.07%		Val Loss: 0.7436	-	Val Acc: 81.64%
Epoch [15/15]		Train Loss: 0.0358	-	Train Acc: 98.68%		Val Loss: 1.1647	-	Val Acc: 78.20%

Best checkpoint saved: best_model/chest_ResNet18.pth (epoch=7, val_acc=95.98%)
Training Complete!

Final Test Results (ResNet18 on chest): Accuracy = 84.46% | Precision (macro) = 0.8866 | Recall (macro) = 0.7970 | Macro F1 = 0.8168

Dataset Used – lesions

Skin lesion dermatoscopy (7 classes, RGB) | Expected minimum test accuracy: 67%

7. lesions – AlexNet

Total epochs trained: 25

Configuration Used:

ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 3
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 25
LEARNING_RATE..... 0.001
NUM_CLASSES..... 7
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 1e-4

Logs:

Epoch [01/25]		Train Loss: 1.0529	-	Train Acc: 65.61%		Val Loss: 1.1957	-	Val Acc: 70.16%
Epoch [02/25]		Train Loss: 0.8855	-	Train Acc: 67.90%		Val Loss: 0.8012	-	Val Acc: 70.41%
Epoch [03/25]		Train Loss: 0.8424	-	Train Acc: 70.00%		Val Loss: 0.7480	-	Val Acc: 73.41%
Epoch [04/25]		Train Loss: 0.8165	-	Train Acc: 70.72%		Val Loss: 0.9198	-	Val Acc: 63.05%
Epoch [05/25]		Train Loss: 0.8063	-	Train Acc: 70.76%		Val Loss: 0.8240	-	Val Acc: 71.16%
Epoch [06/25]		Train Loss: 0.7798	-	Train Acc: 72.01%		Val Loss: 0.6731	-	Val Acc: 75.53%
Epoch [07/25]		Train Loss: 0.7447	-	Train Acc: 72.62%		Val Loss: 0.6804	-	Val Acc: 74.78%
Epoch [08/25]		Train Loss: 0.7338	-	Train Acc: 73.69%		Val Loss: 0.6499	-	Val Acc: 75.91%
Epoch [09/25]		Train Loss: 0.7204	-	Train Acc: 73.98%		Val Loss: 0.6923	-	Val Acc: 74.66%
Epoch [10/25]		Train Loss: 0.7095	-	Train Acc: 74.60%		Val Loss: 0.6509	-	Val Acc: 76.78%
Epoch [11/25]		Train Loss: 0.6972	-	Train Acc: 74.38%		Val Loss: 0.7455	-	Val Acc: 73.41%
Epoch [12/25]		Train Loss: 0.6763	-	Train Acc: 75.61%		Val Loss: 0.6667	-	Val Acc: 75.28%
Epoch [13/25]		Train Loss: 0.6642	-	Train Acc: 75.68%		Val Loss: 0.8356	-	Val Acc: 72.53%
Epoch [14/25]		Train Loss: 0.6372	-	Train Acc: 76.40%		Val Loss: 0.9236	-	Val Acc: 60.67%

Epoch [15/25]	Train Loss: 0.6312	- Train Acc: 76.95%	Val Loss: 1.0745	- Val Acc: 55.93%
Epoch [16/25]	Train Loss: 0.6127	- Train Acc: 77.20%	Val Loss: 0.6772	- Val Acc: 75.28%
Epoch [17/25]	Train Loss: 0.6067	- Train Acc: 77.76%	Val Loss: 0.6645	- Val Acc: 76.65%
Epoch [18/25]	Train Loss: 0.5895	- Train Acc: 78.18%	Val Loss: 0.6799	- Val Acc: 74.28%
Epoch [19/25]	Train Loss: 0.5719	- Train Acc: 78.68%	Val Loss: 0.9693	- Val Acc: 72.53%
Epoch [20/25]	Train Loss: 0.5632	- Train Acc: 79.40%	Val Loss: 0.6156	- Val Acc: 78.28%
Epoch [21/25]	Train Loss: 0.5479	- Train Acc: 80.08%	Val Loss: 0.6048	- Val Acc: 78.78%
Epoch [22/25]	Train Loss: 0.5367	- Train Acc: 80.25%	Val Loss: 1.0047	- Val Acc: 74.78%
Epoch [23/25]	Train Loss: 0.5145	- Train Acc: 81.59%	Val Loss: 1.0500	- Val Acc: 62.92%
Epoch [24/25]	Train Loss: 0.5147	- Train Acc: 81.05%	Val Loss: 1.1849	- Val Acc: 58.55%
Epoch [25/25]	Train Loss: 0.5011	- Train Acc: 81.45%	Val Loss: 0.6626	- Val Acc: 77.15%

Best checkpoint saved: best_model/lesions_AlexNet.pth (epoch=21, val_acc=78.78%)
Training Complete!

Final Test Results (AlexNet on lesions): Accuracy = 76.01% | Precision (macro) = 0.5651 | Recall (macro) = 0.4770 | Macro F1 = 0.4786

8. lesions – VGG16

Total epochs trained: 25

Configuration Used:

```

ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 3
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 25
LEARNING_RATE..... 0.001
NUM_CLASSES..... 7
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 1e-4

```

Logs:

Epoch [01/25]	Train Loss: 1.0747	- Train Acc: 65.63%	Val Loss: 0.9127	- Val Acc: 70.16%
Epoch [02/25]	Train Loss: 0.9660	- Train Acc: 66.60%	Val Loss: 0.8620	- Val Acc: 69.29%
Epoch [03/25]	Train Loss: 0.9424	- Train Acc: 66.32%	Val Loss: 0.8262	- Val Acc: 68.54%
Epoch [04/25]	Train Loss: 0.9072	- Train Acc: 67.01%	Val Loss: 0.8024	- Val Acc: 70.79%
Epoch [05/25]	Train Loss: 0.9077	- Train Acc: 66.83%	Val Loss: 0.7764	- Val Acc: 71.66%
Epoch [06/25]	Train Loss: 0.8819	- Train Acc: 67.62%	Val Loss: 0.7903	- Val Acc: 71.29%
Epoch [07/25]	Train Loss: 0.8882	- Train Acc: 67.43%	Val Loss: 0.8117	- Val Acc: 69.29%
Epoch [08/25]	Train Loss: 0.8679	- Train Acc: 68.01%	Val Loss: 0.7641	- Val Acc: 71.91%
Epoch [09/25]	Train Loss: 0.8726	- Train Acc: 68.37%	Val Loss: 0.8375	- Val Acc: 67.04%
Epoch [10/25]	Train Loss: 0.8446	- Train Acc: 69.64%	Val Loss: 0.7748	- Val Acc: 72.78%
Epoch [11/25]	Train Loss: 0.8310	- Train Acc: 69.57%	Val Loss: 0.7272	- Val Acc: 74.28%
Epoch [12/25]	Train Loss: 0.8144	- Train Acc: 69.82%	Val Loss: 0.7338	- Val Acc: 73.53%
Epoch [13/25]	Train Loss: 0.8125	- Train Acc: 70.27%	Val Loss: 0.7923	- Val Acc: 70.29%
Epoch [14/25]	Train Loss: 0.8022	- Train Acc: 70.50%	Val Loss: 0.8496	- Val Acc: 69.04%
Epoch [15/25]	Train Loss: 0.7924	- Train Acc: 70.61%	Val Loss: 0.7144	- Val Acc: 72.91%
Epoch [16/25]	Train Loss: 0.7932	- Train Acc: 70.61%	Val Loss: 0.9979	- Val Acc: 62.17%
Epoch [17/25]	Train Loss: 0.7909	- Train Acc: 70.76%	Val Loss: 0.6833	- Val Acc: 73.66%
Epoch [18/25]	Train Loss: 0.7714	- Train Acc: 71.47%	Val Loss: 0.7171	- Val Acc: 72.28%
Epoch [19/25]	Train Loss: 0.7749	- Train Acc: 71.79%	Val Loss: 0.7009	- Val Acc: 75.16%
Epoch [20/25]	Train Loss: 0.7584	- Train Acc: 71.83%	Val Loss: 1.5402	- Val Acc: 56.05%
Epoch [21/25]	Train Loss: 0.7425	- Train Acc: 73.23%	Val Loss: 0.8270	- Val Acc: 69.79%
Epoch [22/25]	Train Loss: 0.7476	- Train Acc: 72.51%	Val Loss: 0.7117	- Val Acc: 73.16%
Epoch [23/25]	Train Loss: 0.7246	- Train Acc: 73.10%	Val Loss: 0.8416	- Val Acc: 74.41%
Epoch [24/25]	Train Loss: 0.7221	- Train Acc: 73.57%	Val Loss: 0.6949	- Val Acc: 75.03%
Epoch [25/25]	Train Loss: 0.7133	- Train Acc: 74.27%	Val Loss: 0.8404	- Val Acc: 67.29%

Best checkpoint saved: best_model/lesions_VGG16.pth (epoch=19, val_acc=75.16%)
Training Complete!

[sklearn] UndefinedMetricWarning: precision ill-defined (some classes had 0 predicted samples).

Final Test Results (VGG16 on lesions): Accuracy = 71.22% | Precision (macro) = 0.3895 | Recall (macro) = 0.3053 | Macro F1 = 0.3239

9. lesions – ResNet18

Total epochs trained: 25

Configuration Used:

```

ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 3
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data

```

```
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 25
LEARNING_RATE..... 0.001
NUM_CLASSES..... 7
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 1e-4
```

Logs:

```
Epoch [01/25] | Train Loss: 0.9720 - Train Acc: 66.03% | Val Loss: 0.9265 - Val Acc: 68.41%
Epoch [02/25] | Train Loss: 0.8537 - Train Acc: 68.29% | Val Loss: 0.7988 - Val Acc: 71.79%
Epoch [03/25] | Train Loss: 0.8348 - Train Acc: 69.14% | Val Loss: 0.7086 - Val Acc: 74.03%
Epoch [04/25] | Train Loss: 0.8075 - Train Acc: 70.56% | Val Loss: 0.8540 - Val Acc: 63.92%
Epoch [05/25] | Train Loss: 0.7779 - Train Acc: 70.69% | Val Loss: 0.7274 - Val Acc: 72.53%
Epoch [06/25] | Train Loss: 0.7519 - Train Acc: 72.01% | Val Loss: 0.7053 - Val Acc: 73.16%
Epoch [07/25] | Train Loss: 0.7354 - Train Acc: 72.51% | Val Loss: 0.7191 - Val Acc: 72.16%
Epoch [08/25] | Train Loss: 0.7096 - Train Acc: 73.64% | Val Loss: 0.9580 - Val Acc: 67.29%
Epoch [09/25] | Train Loss: 0.7143 - Train Acc: 73.24% | Val Loss: 0.7481 - Val Acc: 73.03%
Epoch [10/25] | Train Loss: 0.6814 - Train Acc: 74.85% | Val Loss: 0.8472 - Val Acc: 69.16%
Epoch [11/25] | Train Loss: 0.6762 - Train Acc: 75.35% | Val Loss: 0.7199 - Val Acc: 71.41%
Epoch [12/25] | Train Loss: 0.6628 - Train Acc: 75.35% | Val Loss: 0.6290 - Val Acc: 76.65%
Epoch [13/25] | Train Loss: 0.6530 - Train Acc: 75.92% | Val Loss: 0.7214 - Val Acc: 72.41%
Epoch [14/25] | Train Loss: 0.6519 - Train Acc: 75.52% | Val Loss: 0.6432 - Val Acc: 76.03%
Epoch [15/25] | Train Loss: 0.6401 - Train Acc: 76.24% | Val Loss: 0.7095 - Val Acc: 72.41%
Epoch [16/25] | Train Loss: 0.6296 - Train Acc: 76.72% | Val Loss: 0.6952 - Val Acc: 74.78%
Epoch [17/25] | Train Loss: 0.6165 - Train Acc: 76.96% | Val Loss: 0.6660 - Val Acc: 77.03%
Epoch [18/25] | Train Loss: 0.6096 - Train Acc: 76.95% | Val Loss: 0.6209 - Val Acc: 77.15%
Epoch [19/25] | Train Loss: 0.5981 - Train Acc: 77.86% | Val Loss: 0.6167 - Val Acc: 77.65%
Epoch [20/25] | Train Loss: 0.5887 - Train Acc: 77.89% | Val Loss: 0.8091 - Val Acc: 70.79%
Epoch [21/25] | Train Loss: 0.5802 - Train Acc: 78.30% | Val Loss: 0.6184 - Val Acc: 78.53%
Epoch [22/25] | Train Loss: 0.5553 - Train Acc: 79.16% | Val Loss: 0.6607 - Val Acc: 74.91%
Epoch [23/25] | Train Loss: 0.5361 - Train Acc: 79.94% | Val Loss: 0.6097 - Val Acc: 77.78%
Epoch [24/25] | Train Loss: 0.5393 - Train Acc: 79.62% | Val Loss: 0.7076 - Val Acc: 75.16%
Epoch [25/25] | Train Loss: 0.5107 - Train Acc: 81.05% | Val Loss: 0.9810 - Val Acc: 63.80%
Best checkpoint saved: best_model/lesions_ResNet18.pth (epoch=21, val_acc=78.53%)
Training Complete!
```

[sklearn] UndefinedMetricWarning: precision ill-defined (some classes had 0 predicted samples).

Final Test Results (ResNet18 on lesions): Accuracy = 76.46% | Precision (macro) = 0.6136 | Recall (macro) = 0.4205 | Macro F1 = 0.4719

Dataset Used – orgs

Organ images (11 classes, grayscale) | Expected minimum test accuracy: 83%

10. orgs – AlexNet

Total epochs trained: 15

Configuration Used:

```
ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 1
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 11
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.8939 - Train Acc: 68.87% | Val Loss: 0.4486 - Val Acc: 82.88%
Epoch [02/15] | Train Loss: 0.3669 - Train Acc: 88.23% | Val Loss: 0.4018 - Val Acc: 87.43%
Epoch [03/15] | Train Loss: 0.2964 - Train Acc: 90.30% | Val Loss: 0.4067 - Val Acc: 87.76%
Epoch [04/15] | Train Loss: 0.2334 - Train Acc: 92.26% | Val Loss: 0.1587 - Val Acc: 94.27%
Epoch [05/15] | Train Loss: 0.2069 - Train Acc: 93.15% | Val Loss: 0.2430 - Val Acc: 92.25%
Epoch [06/15] | Train Loss: 0.1724 - Train Acc: 94.26% | Val Loss: 0.2111 - Val Acc: 91.86%
Epoch [07/15] | Train Loss: 0.1439 - Train Acc: 95.13% | Val Loss: 0.1677 - Val Acc: 94.86%
Epoch [08/15] | Train Loss: 0.1488 - Train Acc: 94.99% | Val Loss: 0.1683 - Val Acc: 94.14%
Epoch [09/15] | Train Loss: 0.1360 - Train Acc: 95.57% | Val Loss: 0.3312 - Val Acc: 89.06%
Epoch [10/15] | Train Loss: 0.1337 - Train Acc: 95.65% | Val Loss: 0.0980 - Val Acc: 96.03%
```

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
[11/15]	0.1275	95.78%	0.2609	94.08%
[12/15]	0.1035	96.56%	0.5707	87.37%
[13/15]	0.1141	96.32%	1.6973	77.60%
[14/15]	0.1044	96.55%	0.2073	92.90%
[15/15]	0.1002	96.94%	0.1662	94.21%

Best checkpoint saved: best_model/orgs_AlexNet.pth (epoch=10, val_acc=96.03%)
Training Complete!

Final Test Results (AlexNet on orgs): Accuracy = 90.07% | Precision (macro) = 0.8894 | Recall (macro) = 0.8856 | Macro F1 = 0.8855

11. orgs – VGG16

Total epochs trained: 15

Configuration Used:

```

ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 1
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 11
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 0

```

Logs:

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
[01/15]	1.3792	49.57%	1.0464	61.65%
[02/15]	0.6527	75.22%	0.6058	78.58%
[03/15]	0.5041	80.48%	0.3726	84.05%
[04/15]	0.4022	84.26%	0.3266	86.52%
[05/15]	0.3426	87.14%	0.3445	86.00%
[06/15]	0.3499	87.55%	0.2428	90.56%
[07/15]	0.2900	89.56%	0.2845	89.97%
[08/15]	0.2303	91.90%	0.3085	89.52%
[09/15]	0.2082	93.23%	0.3986	88.28%
[10/15]	0.3449	89.00%	0.1844	93.55%
[11/15]	0.1889	93.63%	0.1514	94.60%
[12/15]	0.1877	93.93%	0.1524	94.66%
[13/15]	0.1630	94.74%	0.1758	94.14%
[14/15]	0.1272	95.72%	0.1396	95.44%
[15/15]	0.1163	96.31%	0.1480	94.79%

Best checkpoint saved: best_model/orgs_VGG16.pth (epoch=14, val_acc=95.44%)
Training Complete!

Final Test Results (VGG16 on orgs): Accuracy = 88.90% | Precision (macro) = 0.8812 | Recall (macro) = 0.8691 | Macro F1 = 0.8722

12. orgs – ResNet18

Total epochs trained: 15

Configuration Used:

```

ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 1
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 11
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 0

```

Logs:

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
[01/15]	0.6646	75.92%	0.5742	81.32%
[02/15]	0.3328	88.06%	0.6068	80.21%
[03/15]	0.2415	91.37%	0.1964	92.90%
[04/15]	0.1867	93.49%	0.2044	92.84%
[05/15]	0.1594	94.33%	0.2323	91.54%
[06/15]	0.1257	95.46%	0.2882	91.08%

Epoch [07/15] | Train Loss: 0.1260 - Train Acc: 95.52% | Val Loss: 0.1832 - Val Acc: 93.95%
 Epoch [08/15] | Train Loss: 0.0910 - Train Acc: 96.81% | Val Loss: 0.1936 - Val Acc: 92.90%
 Epoch [09/15] | Train Loss: 0.1117 - Train Acc: 96.02% | Val Loss: 0.0998 - Val Acc: 96.16%
 Epoch [10/15] | Train Loss: 0.0885 - Train Acc: 97.01% | Val Loss: 0.1554 - Val Acc: 94.99%
 Epoch [11/15] | Train Loss: 0.0644 - Train Acc: 97.75% | Val Loss: 0.1134 - Val Acc: 95.96%
 Epoch [12/15] | Train Loss: 0.0489 - Train Acc: 98.25% | Val Loss: 0.1877 - Val Acc: 94.08%
 Epoch [13/15] | Train Loss: 0.0668 - Train Acc: 97.67% | Val Loss: 0.0999 - Val Acc: 96.09%
 Epoch [14/15] | Train Loss: 0.0460 - Train Acc: 98.45% | Val Loss: 0.0729 - Val Acc: 97.14%
 Epoch [15/15] | Train Loss: 0.0408 - Train Acc: 98.58% | Val Loss: 0.1739 - Val Acc: 94.79%
 Best checkpoint saved: best_model/orgs_ResNet18.pth (epoch=14, val_acc=97.14%)
 Training Complete!

Final Test Results (ResNet18 on orgs): Accuracy = 92.00% | Precision (macro) = 0.9109 | Recall (macro) = 0.9088 | Macro F1 = 0.9078

Consolidated Summary – All Permutations

Dataset	Model	Accuracy (%)	Precision	Recall	Macro F1
cells	AlexNet	95.15	0.947	0.944	0.945
cells	VGG16	96.78	0.965	0.969	0.967
cells	ResNet18	96.35	0.960	0.964	0.961
chest	AlexNet	86.38	0.905	0.820	0.841
chest	VGG16	82.21	0.889	0.763	0.782
chest	ResNet18	84.46	0.887	0.797	0.817
lesions	AlexNet	76.01	0.565	0.477	0.479
lesions	VGG16	71.22	0.390	0.305	0.324
lesions	ResNet18	76.46	0.614	0.420	0.472
orgs	AlexNet	90.07	0.889	0.886	0.885
orgs	VGG16	88.90	0.881	0.869	0.872
orgs	ResNet18	92.00	0.911	0.909	0.908